

Access Database Basics

Lesson 28 — Databases, Tables and Records

JKSSB Computer Awareness

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What Is a Database?

- ▶ A **database** is an organised collection of related data.
- ▶ The data is stored so that it can be **searched, updated, and retrieved** easily.
- ▶ Example: a school's list of students, their classes, and their marks kept together as related information.
- ▶ In a computer, a database is managed by special software, not kept as loose files.

The one idea

A database is the **collection of data**; the software that manages it is a separate thing.

Why Use a Database Instead of a Spreadsheet?

- ▶ A spreadsheet is built for calculation on a simple grid.
- ▶ A database is built for **storing and linking large amounts of related data**.
- ▶ A database can hold many tables, relate them, and answer queries across them.
- ▶ For a big, growing record system, a **database** keeps data consistent and avoids duplication.

Keep it clear

Excel is a spreadsheet; Access is a database program. The two are not the same tool.

Database vs DBMS (Near-Neighbour)

Point	Database	DBMS
What it is	The organised collection of data	The software that manages the data
Nature	Data	Program
Example	A school's records	MS Access, Oracle, MySQL

The database is the data; the DBMS is the tool. Do not swap them.

What a DBMS Does

- ▶ **DBMS** stands for **Database Management System**.
- ▶ It **stores** data and controls how it is kept on disk.
- ▶ It lets users **insert, update, delete, and retrieve** records.
- ▶ It **controls access** so that only allowed users can read or change data.
- ▶ It keeps the data **consistent** and safe from corruption.

Full form to memorise

DBMS = Database Management System. An **RDBMS** is a relational DBMS (data stored in related tables).

Microsoft Access in the Office Suite

- ▶ **Microsoft Access** is the database program of the **Microsoft Office** suite.
- ▶ It is a **relational database management system** (RDBMS).
- ▶ In the suite, Word handles text, Excel handles spreadsheets, PowerPoint handles presentations, and **Access handles databases**.
- ▶ Access stores tables, queries, forms, and reports inside one database file.

Exam point

Access is a DBMS, not a spreadsheet. It is part of the MS Office family.

The Objects in an Access Database

Object	Job
Table	Stores the actual data in rows and columns
Query	Asks a question and retrieves selected data
Form	Provides a screen to enter and view data
Report	Presents data in a printable layout

Data lives in tables; the other objects work on that data.

The Table: Rows and Columns

- ▶ A **table** is the basic place where data is stored.
- ▶ Data is arranged in **rows and columns**.
- ▶ Each **column** is one **field** — one kind of information.
- ▶ Each **row** is one **record** — one complete entry.
- ▶ A table therefore holds **many records**, each with the same set of fields.

The building block

Everything in an Access database starts with the **table**.

Field vs Record vs Table (Near-Neighbour)

Term	Meaning	Position in a table
Field	One kind of information	A column
Record	One complete entry	A row
Table	A collection of related records	The whole grid

Trap alert

Do not mix these up: a **field** is a column, a **record** is a **row**, and a **table** is the whole collection.

Data Types in Access

- ▶ Every field has a **data type** — the kind of value it can hold.
- ▶ **Short Text** — names, codes, and other text.
- ▶ **Number** — values used in calculations.
- ▶ **Date/Time** — dates and times.
- ▶ **Yes/No** — only True or False (a Boolean value).
- ▶ **AutoNumber** — a unique number added automatically for each new record.

Why it matters

A field can hold only values that match its data type, which keeps the data clean.

Field Properties

- ▶ A **field property** is a setting that describes how a field behaves.
- ▶ **Field Size** limits how much a field can hold.
- ▶ **Default Value** fills in a value automatically.
- ▶ **Required** — if set to Yes, the field **cannot be left blank**.
- ▶ **Input Mask** fixes the pattern of entry, such as a phone number.

Input Validation

- ▶ **Validation** means checking that entered data is sensible and allowed.
- ▶ A **Validation Rule** states the condition the value must meet.
- ▶ Example rule: **Between 0 And 100** — only values in that range are accepted.
- ▶ **Validation Text** is the message shown when the rule is violated.

Worked example

Set a Validation Rule of Between 0 And 100 on a Marks field. If someone types 150, Access rejects it and shows the Validation Text.

The Primary Key

- ▶ A **primary key** is a field that **uniquely identifies** each record.
- ▶ Its value is **different for every record** and is never left blank.
- ▶ Example: a **Roll Number** identifies each student in a class.
- ▶ An **AutoNumber** field is often used as a ready-made primary key.

Trap alert

A primary key **cannot** contain duplicate values. If two records share the same key value, Access will not allow it.

Indexes

- ▶ An **index** is a special structure that makes searching and sorting a table **faster**.
- ▶ It works like the index of a book, pointing straight to the right place.
- ▶ A primary key is **automatically indexed**.
- ▶ An index takes extra space and can slightly slow down adding data, so it is used only where needed.

Remember

Index = speed of **search**; it does not change the data itself.

Worked Example: A Students Table

RollNo	Name	Marks	Passed
1	Ayaan	78	Yes
2	Bilal	45	No
3	Chitra	91	Yes

- ▶ **RollNo** is the primary key (unique for each student).
- ▶ **Name** is Short Text; **Marks** is Number; **Passed** is Yes/No.
- ▶ Each row is one **record**; each column is one **field**.

Trap Alert: Common Misreadings

- ▶ A **database** is the data; a **DBMS** is the software. (TRAP-style)
- ▶ A **field** is a column; a **record** is a row.
- ▶ **Validation** checks the value; an **index** speeds up search.
- ▶ A **primary key** is unique; it cannot contain duplicates.
- ▶ Access is part of **MS Office**, not a spreadsheet.

Lesson 28: Recall

- ▶ A database is an organised collection of related data; a DBMS is the software that manages it.
- ▶ Microsoft Access is the RDBMS of the MS Office suite.
- ▶ The table stores data in **fields** (columns) and **records** (rows).
- ▶ Data types include Short Text, Number, Date/Time, Yes/No, and AutoNumber.
- ▶ Validation Rules and Validation Text check values; Required stops blank entries.
- ▶ A primary key uniquely identifies a record; an index speeds up searching.